



# LAB 5, FALL 2026

## CONCEPTS FOR LAB 5

- Vectors
- OpenMP
- Parallel Programming
- Quicksort

## THIS IS NOT A PAIRED PROGRAMMING ASSIGNMENT

All students are to complete this assignment on their own. You all can talk to each other but you definitely shouldn't have the same code! If you do, that is considered cheating for this assignment (refer to syllabus academic integrity policy).

## LAB 5 POLICIES

### Generative AI Policy for Lab 5

You are **NOT** allowed to use GitHub Copilot, OpenAI Codex, or any other AI generation tool for **code generation**. Therefore, if you currently have GenAI enabled within your IDE, you need to **disable it for completing this lab**. You should try to implement the sorting algorithms yourself without outside help so that you can learn these algorithms and how to implement them.

You may use Generative AI to aid in **troubleshooting your code** without uploading the given source file. Do not upload the assignment directions and ask it to generate the solution or parts of the solution for you.

**Uploading the assignment or given .cpp file to generate code, fix the given code, or enabling code generation/completion within VSCode will result in an academic integrity violation charge.**

### What you Can't DO/Use in this lab

**If you do any of the following, you will receive a 0 on some or all of the assignment and possibly get charged with an academic integrity violation charge (this is very serious).**

1. You may not use more advanced code or syntax in your lab assignment **that has not been taught** in CSC 1310 class **without referencing your source**. How do you reference your source? By providing a link/URL in a multi-line comment in the section of code where you use the advanced code. **Otherwise, this will be considered plagiarism.**
2. You should be able to explain every line of code that is in your submission. Look at syllabus to review what you may not (ever) use in your programming & lab assignments in CSC 1310
3. You **may not use generative AI (genAI)** to generate code, but you can use it to help in troubleshooting problems as described in the Generative AI Policy above.
4. You may not share your code with anyone.

## GIVEN FILES

You will be given part of the program and several different .txt files to test your program.

- **vector examples (vector4.cpp, vector5.cpp, vector6.cpp)**
- **parallelQuickSort.cpp**
- **movies.txt**
- **lab5\_given.cpp**

## DIRECTIONS

1. Review **vector4.cpp**, **vector5.cpp**, and **vector6.cpp**.
2. Download **lab5\_given.cpp**, rename it to **lab5.cpp**, and complete the program. This is just to practice with vectors.
3. Review **parallelQuickSort.cpp**
4. Compile & run parallelQuickSort.cpp. Take a screenshot of parallelQuickSort running on your computer and name it **Lab5RunProgProof**.
5. Begin (or continue) working on Program 2.

## DON'T FORGET TO FILL OUT THE LAB REPORT

**After** you have completed this assignment, fill out the lab report and submit your lab report in your zip file along with your source files. Name your screenshot **lab5ReportProof**.

**Lab Report Link:** [https://tntech.co1.qualtrics.com/jfe/form/SV\\_0iJcCTCBgguPDpQ](https://tntech.co1.qualtrics.com/jfe/form/SV_0iJcCTCBgguPDpQ)

Create a **zip** file named **username\_lab5.zip** containing the following files and upload it to Canvas in the **Lab 5** assignment. Replace username with your TTU. Example: **acrockett43\_lab5.zip**

- **Lab5.cpp**
- **Lab5RunProgProof**
- **Lab5ReportProof**

## SAMPLE OUTPUT OF RUNNING LAB5.CPP

```
Welcome to Crockett Farms Pumpkin Festival!
Please enter each activity you did at the festival and score the activities.

Then, rate the activity from 1 to 100 where 1 means the activity was buns
and 100 means it was your favorite activity you have ever done in your life.

Enter "done" to quit entering activities.

Activity: Pick a Pumpkin
Rating: 75

Activity: Pumpkin Carving
Rating: 90

Activity: Hayride
Rating: 70

Activity: Face Painting
Rating: 30

Activity: Apple Bobbing
Rating: 0

Activity: Haunted Corn Maze
Rating: 100

Activity: done
```

Activities Entered:

Fall Activities:

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Pick a Pumpkin: 75  
Pumpkin Carving: 90  
Hayride: 70  
Face Painting: 30  
Apple Bobbing: 0  
Haunted Corn Maze: 100

Activities Sorted by Rating:

Fall Activities:

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Haunted Corn Maze: 100  
Pumpkin Carving: 90  
Pick a Pumpkin: 75  
Hayride: 70  
Face Painting: 30  
Apple Bobbing: 0